AMERICAN INTERNATIONAL UNIVERSTY-BANGLADESH

Faculty Of Science and Technology

Assignment Title: **Group Assignment**

Course Name: Physics 2 [PHY1203]; Section: H

Course Teacher: MD. MASUD PARVEZ; Group: **C**

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No	Name	ID	Program	Math
1.	HABIB, MD. IMTIAZ	19-39389-1	BSc CSE	3, 11, 12, 16, 18
2.	KATAB, TANJIL BIN	19-39966-1	BSc EEE	4, 7, 10, 15, 19
3.	ISLAM, TAJUL	19-39847-1	BSc CSE	1, 5, 8, 14, 17
4.	MUBIN, SAKIB HASAN	19-40715-1	BSc CSE	2, 6, 9, 13, 20

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1] (a) As given

$$t = \frac{T}{2}$$

$$\Rightarrow T = 2t$$

$$= 2 \times 0.25$$

$$= 0.5s$$

(b) $$f = \frac{1}{T}$$

$$= \frac{1}{0.5}$$

$$= 2Hz$$

(c) Amplitude of the motion $= \frac{d}{2}$

$$= \frac{36}{2}$$

$$= 18cm$$ (Ans

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2. Given,
The displacement at time, $t = 2$ sec is given by

$$x = (6.0\,m)\cos[(3\pi)t + \pi/3]$$

(a) Here, $t = 2$ sec

$\therefore$ displacement $x_t = (6.0\,m)\cos[(3\pi)t + \pi/3]$

$= (6.0\,m)\cos[(3\pi)(2s) + \frac{\pi}{3}]$

$= (6\,m)\cos(6\pi + \frac{\pi}{3})$

$= (6\,m)(\frac{1}{2})$

$= 3\,m$

Note that, in all ~~earl~~ calculations, we convert π inside sin or cos from rad to degree where the angle by 57.28°/rad.

(b) Given, $x = (6.0\,m)\cos[(3\pi)t + \pi/3]$ —— (1)

Take the 1st derivative for eqn (1) we can get the velocity, $v = \frac{dx}{dt}$

$= \frac{d}{dt}(6\,m)\cos[(3\pi)t + \frac{\pi}{3}]$

$= -6(3\pi)\sin(3\pi t + \frac{\pi}{3})$ —— (2)

$(\sin\theta)$

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Now, put value for $t = 2$ sec into eqn (2) to get the velocity, $v = -6(3\pi)\sin(3\pi t + \pi/3)$

$$= -6(3\pi)\sin(6\pi + \pi/3)$$

$$\boxed{= -49\ ms^{-1}} \text{ (Ans)}$$

(c) from answer (b) we get eqn (2) and by derivative it we get the acceleration

$$a = \frac{d^2x}{dt^2}$$

$$= \frac{d^2}{dt^2}\left[-6(3\pi)^2 \sin(3\pi t + \frac{\pi}{3})\right]$$

$$= -6(3\pi)^2 \cos(3\pi t + \frac{\pi}{3}) \quad \text{——— (3)}$$

Now, put the value for $t = 2$ sec in eqn (3) to get the acceleration, $a = -6(3\pi)^2 \cos(3\pi t + \frac{\pi}{3})$

$$= -6(3\pi)^2 \cos(6\pi + \frac{\pi}{3})$$

$$\boxed{= -267\ ms^{-2}} \text{ (Ans)}$$

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(d) The displacement $x(t)$ of a particle from its equillibrium position is described by

$$\boxed{x = x_m \cos(\omega t + \phi)} \quad \text{—— (4)}$$

From answer (b) we get eqn (1) and comparing (1) and (4) we can get, the phase of the motion

$$(\omega t + \phi) = (3\pi)t + \pi/3$$

Now, at $t = 2$ sec

$$(\omega t + \phi) = 3\pi t + \pi/3 = (3\pi \times 2 + \pi/3)$$

$$\boxed{= 20 \text{ rad}} \quad \text{(Ans)}$$

$\therefore$ Angular frequency

$$\boxed{\omega = 3\pi}$$

(e) The angular frequency ω is related to the period and frequency of the motion by

$$\omega = 2\pi f$$

$$\Rightarrow f = \frac{\omega}{2\pi}$$

$$\Rightarrow f = \frac{3\pi}{2\pi} = \boxed{1.5 \text{ Hz}}$$

(Ans)

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(f) We know,

$$T = \frac{1}{f}$$

$$= \frac{1}{1.5 \text{ Hz}} \quad [f = 1.5 \text{ Hz}]$$

$$= 0.67 \text{ sec} \quad \text{(Ans)}$$

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3 Given, $m = 0.5\ kg$

The amplitude $x_m = 35\ cm$

$t = 0.5_s$

(a) The period will be,

$$T = t = 0.5_s$$

(Ans:)

(b) The frequency,

$$f = \frac{1}{T} = \frac{1}{0.5}$$

$$\therefore f = 2\ Hz$$

(Ans:)

(c) The angular frequency,

$$\omega = 2\pi f$$
$$= 2 \times 3.1416 \times 2$$
$$= 12.56\ rad\,s^{-1}$$
$$= 4\pi\ rad\,s^{-1}$$

(Ans:)

(d) Spring constant, $F = ma$

$$\Rightarrow Kx = m\omega^2 x$$
$$\Rightarrow K = m\omega^2$$
$$= 0.5 \times (12.56)^2$$
$$= 78.96\ N\,m^{-1}$$

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(e) Maximum speed,

$$V_{max} = \pm \omega x_m$$
$$= \pm 12.56 \times 35 \times 10^{-2}$$
$$= 4.39 \text{ ms}^{-1}$$ (Ans:)

(f) Magnitude of the maximum force,

$$|F_{max}| = m\omega^2 x_m$$
$$= 0.5 \times (12.56)^2 \times 35 \times 10^{-2}$$
$$= 27.63$$ (Ans:)

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9.

Given,

$m = 2\ kg$

$k = 100\ Nm^{-1}$

$x = 0.129\ m$

$V = 3.915\ ms^{-1}$

(a) $k = m\omega^2$

$\therefore \omega = \sqrt{\frac{k}{m}}\ rads^{-1}$

$= \sqrt{\frac{100}{2}}\ rads^{-1}$

$= 7.07\ rads^{-1}$

$V = \omega\sqrt{A^2 - x^2}$

$\Rightarrow V^2 = \omega^2 (A^2 - x^2)$

$\therefore A^2 = \frac{V^2}{\omega^2} + x^2$

$\therefore A = \sqrt{\frac{V^2}{\omega^2} + x^2}$

$= \sqrt{\left(\frac{3.915}{7.07}\right)^2 + 0.129^2}$

$= 0.5\ m.$ (Ans.)

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(b) $x = A \sin(\omega t + \delta)$

when, $t = 1$, $x = 0.129$

$0.129 = 0.5 \times \sin(7.07 + \delta)$

$\Rightarrow \sin(7.07 + \delta) = 0.258$

$\therefore \delta = 7.88^\circ$

$\therefore$ position at $t = 0$

$x = A \sin\delta$

$= 0.5 \times \sin 7.88$

$= 0.0685 \text{ m}$. (Ans,)

(a) $V_o = \omega\sqrt{A^2 - x^2}$

$= 7.07 \sqrt{(0.5)^2 - (0.0685)^2}$

$= 3.5 \text{ ms}^{-1}$. (Ans,)

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5/(a)

$a = -\omega^2 x$

$\therefore \omega = \sqrt{\frac{a}{x}}$

$= \sqrt{\frac{123}{0.129}} = 30.88 \text{ rad s}^{-1}$

Given

K = 400 N

a = -123

x = 0.12

$\omega = 2\pi f$

$\therefore f = \frac{\omega}{2\pi} = \frac{30.88}{2\pi} = 4.91 \text{ Hz}$

(b)

$K = m\omega^2$

$\therefore m = \frac{K}{\omega^2} = \frac{400}{(30.88)^2} = 0.419 \text{ kg}$

(C)

$A = \sqrt{\frac{v^2}{\omega^2} + x} = \sqrt{\frac{13.6}{30.88} + (0.129)^2}$

$= 0.46 \text{m}$

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(6)

(a) The period of the motion —

$$T = \frac{1}{f} = \frac{1}{6} = \boxed{0.167 \text{ se}}$$ (Ans)

(b) The angular frequency —

$$\omega = 2\pi f = 2\pi(6) = \boxed{37.7 \text{ rad/s}}$$

(Ans)

(c) The mass of the body —

$$\omega = \sqrt{\frac{K}{m}} \Rightarrow m = \frac{K}{\omega^2}$$

$$\Rightarrow m = \frac{120}{37.7} = \boxed{0.0844 \text{ kg}}$$

(Ans)

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(a) amplitude, $A = \sqrt{x_0^2 + \frac{V_0 \dot{x}}{\omega^2}}$

$= \sqrt{x_0^2 + \frac{m V_0^2 x}{k}}$

$= \sqrt{0^2 + \frac{2 \times (12)^2}{300}}$

$= 0.98$ m. (Ans.)

(b) $x = A \cos(\omega t - \phi)$

when $t = 0$, $x = x_0$, $x_0 = A \cos\phi$

for, $\phi = 0$; $x_0 = A$

for $\phi = \pi$; $x_0 = -A$

for $\phi = \pi/2$; $x_0 = 0$

$\therefore \phi = \frac{\pi}{2}$

(c) $x = A \cos(\omega t - \phi)$

$= 0.98 \cos(12.25 t + \frac{\pi}{2})$

Or, $x = -0.98 \sin(12.25 t)$

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Given

$y_m = 6$ $\quad \omega = 600$ rad/s

$y_1 = 2.\cdot m$ $\quad y_2 = 2.\cdot m$

equ^{ns}: $2 = 6\sin(kx + 600t_1 + \phi)$

$-2 = 6\sin(kx + 600t_2 + \phi)$

$\sin^{-1}$ in both equ^{n}6:

$$kx + 600t_1 + \phi = \frac{\pi}{180}\sin^{-1}\left(\frac{1}{3}\right)$$

$$kx + 600t_2 + \phi = \frac{\pi}{180}\sin^{-1}\left(-\frac{1}{3}\right)$$

Substract both:

$$600(t_1 - t_2) = \frac{2\pi}{180}\sin^{-1}\left(\frac{1}{3}\right)$$

$$\therefore \Delta t = (t_1 - t_2) = 0.0011\,s$$

(Ans)

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(9) Given, $\omega = 110$ rad/s, $\lambda = 1.8$ m

(a) The angular wave number

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{1.8} = \boxed{3.48 \text{ m}^{-1}} \text{ (Ans)}$$

(b) The speed of the wave,

$$v = \lambda f = \frac{\lambda \omega}{2\pi} = \frac{1.8 \times 110}{2\pi}$$

$$\boxed{= 31.51 \text{ m/s}} \text{ (Ans)}$$

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10.

(a) $\frac{T}{4} = 0.17$

$\therefore T = 0.68 \text{ s}.$

(Ans.)

(b) $f = \frac{1}{T}$

$= \frac{1}{0.68}$

$= 1.47 \text{ Hz}.$ (Ans.)

(c) $v = f\lambda$

$= 1.47 \times 1.4$

$= 2.058 \text{ ms}^{-1}$ (Ans.)

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11 (a) $\omega = \frac{v_m}{y_m} = \frac{16}{4\times10^{-2}} = 400 \text{ rad/s}$

$f = \frac{\omega}{2\pi} = \frac{400}{2\pi} = 63.66 \text{ Hz}$

(b) Given, $v = 80 \text{ m/s}$

$\therefore \lambda = \frac{v}{f}$

$= \frac{80}{63.66}$

$= 1.25 \text{ m/s}$

(c) Given equation, $y(x,t) = y_m \sin(kx \pm \omega t + \varphi)$

Substituting by $y = 4 \text{ cm}$ at $x = 0$, $t = 0$

$\therefore y_m = 0.04 \text{ m}$

(d) $k = \frac{2\pi}{\lambda}$

$= \frac{2\pi}{1.25} = 5 \text{ rad/s}$

(e) The angular velocity, $\omega = \frac{v_m}{y_m}$

$\Rightarrow \omega = \frac{16}{0.04}$

$\therefore \omega = 400 \text{ rad/s}$

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(f) $y = 4$ cm at $x = 0$, $t = 0$ therefore,

$$0.04 = 0.04 \sin(5(0) \pm 400(t) + \varphi)$$

$$\Rightarrow 1 = \sin(\varphi)$$

$$\Rightarrow \varphi = \sin^{-1}(1)$$

$$\therefore \varphi = \pi/2 \quad //$$

$$\therefore Y = 0.04 \sin\left(5x \pm 400(t) + \frac{\pi}{2}\right)$$

(g) The correct choice of sign in front of (t) is negative (−) because the wave is shown travelling to positive direction of x-axis.

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Given, $y = 6 \sin(0.020\pi x + 4.0\pi t)$

$\therefore t = 0.26\,s$

$x = 3.5\,cm$

By comparing the given equation to the general wave function form, $y = y_m \sin(kx - \omega t + \phi)$

(a) Amplitude, $y_m = 6\ cm$ (Ans:)

(b) $k = \frac{2\pi}{\lambda}$

$\Rightarrow 0.02\pi = \frac{2\pi}{\lambda}$

$\Rightarrow \lambda = \frac{2\pi}{0.02\pi}$

$\therefore \lambda = 100\ cm$ (Ans:)

(c) $\omega = 2\pi f = 4\pi$

$\Rightarrow f = \frac{4\pi}{2\pi}$

$\therefore f = 2\ Hz$ (Ans:)

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(d) The speed, $\Rightarrow v = f\lambda$

$\Rightarrow v = 2 \times 100$

$= 200 \text{ cm/s}$ (Ans:)

(e) The direction of propagation of the wave is in the negative (-ve) direction of the x-axis as we got a positive (+ve) sign in the given equation. (Ans:)

(f) The maximum transverse speed,

$v_m = \omega y_m$

$= 4\pi \times 6$

$= 24\pi = 75.39 \text{ cm/s}$ (Ans:)

(g) Given, $t = 0.26 \text{ s}$; $x = 3.5 \text{ cm}$

$\therefore y = 6 \sin (0.02\pi \times 3.5 + 4\pi \times 0.26)$

Therefore, $y = -2 \text{ cm}$ (Ans:)

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(13) (a) The equation of sinusoidal wave is given by, $y = A \sin(Kx - \omega t)$ traveling in the x-direction — (1)

Comparing Eqn (1) with one in the question, we get

Amplitude → $\boxed{y_{max} = 0.00327 \text{ m}}$ (Ans)

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Comparison of the two wave eqn gives

wavelength, $\lambda = \frac{2\pi}{k} = \frac{2\pi}{72.1} = \boxed{8.72 \text{ cm}}$ (Ans)

Period, $T = \frac{2\pi}{\omega} = \frac{2\pi}{2.72} = \boxed{2.31 \text{ sec}}$ (Ans)

Frequency, $f = \frac{\omega}{2\pi} = \frac{2.72}{2\pi} = \boxed{0.433 \text{ Hz}}$ (Ans)

Velocity of the wave ⟵

$V = \lambda f = (0.087 \times 0.433)$

$= 0.038 \text{ m/s}$

$= \boxed{3.8 \text{ cm/s in the } (+x)\text{-direction}}$ (Ans)

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(ii) Given, $x = 22.5\text{cm}$ and $t = 18.9\ \text{sec}$

$$Y = 0.00327 \sin(72.1 \times 0.225 - 2.72 \times 18.9)$$

$$= 0.00327 \sin(16.22 - 51.41)$$

$$= 0.00327 \sin(-35.19\ \text{radians})$$

$$= 0.00327 \times 0.59$$

$$= 0.00193\ \text{m} = \boxed{1.93\ \text{mm}}\ \text{(Ans)}$$

$$\boxed{= 1.93 \times 10^{-3}\ \text{m}}\ \text{(Ans)}$$

~~(iii)~~ (iv) $a = -\omega^2 [A \sin(kx - \omega t)]$

$$a = -\omega^2 \times y$$

$$\therefore a = -(2.72)^2 \times 1.92 \times 10^{-3}\ \text{ms}^{-2}$$

$$\boxed{= -0.014\ \text{ms}^{-2}}\ \text{(Ans)}$$

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(13) (iii) Given, $y = A \sin(kx - \omega t)$

$$v = -\omega A \cos(kx - \omega t)$$

$$= -2.72 \times 0.00327 \cos(72.1 \times 0.225 - 2.72 \times 18.9)$$

$$\boxed{= 7.197 \times 10^{-3} \ ms^{-1}} \quad \text{(Ans)}$$

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14| (a) As we know;

$$V = f\lambda$$

$$\Rightarrow \lambda = \frac{V}{f}$$

$$= \frac{350}{500}$$

$$= 0.7\,m$$

$$\sigma = \frac{2\pi}{\lambda} \times \Delta x$$

$$\Rightarrow \Delta x = \frac{\sigma\lambda}{2\pi} = \frac{\frac{\pi}{3} \times 0.7}{2\pi}$$

$$= 0.116\,m \quad \text{(Ans)}$$

(b) we know

$$\sigma = \frac{\Delta t}{T} \times 2\pi$$

$$= \frac{0.001}{0.002} \times 2\pi$$

$$= \pi\ rad$$

$$T = \frac{1}{f} = \frac{1}{500}$$

$$= 0.002s$$

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$$P_{avg} = \frac{1}{2} \mu V \omega^2 y_m^2$$

$$V = \sqrt{\frac{T}{\mu}}$$

We know,

$$\omega = 2\pi f$$

$$\therefore P_{avg} = \frac{1}{2} \sqrt{\mu} + (2\pi f)^2 \; y_m^2.$$

the mass per unit length of the string

$$\mu = \frac{m}{l} = \frac{0.26}{2.7}$$

$$= 0.096 \text{ kg/m}.$$

the wave frequency,

$$\therefore 85 = \frac{1}{2} \sqrt{36 \times 0.096} \; (2\pi f)^2 \times (0.0077)^2$$

$$\therefore f = 197.6 \text{ Hz}. \quad \text{(Ans.)}$$

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16 Given, $\mu = 525$ g/m

$v = 9.26$ ms^{-1}

$y_m = 8.5$ mm

$$\omega = 2\pi f$$
$$= 2\pi \times (120)$$
$$= 754 \text{ rad/s}$$

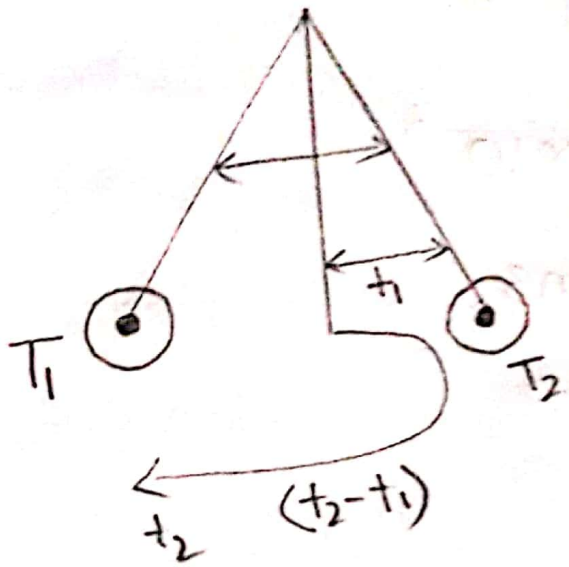
$$P_{avg} = \frac{1}{2} \mu v \omega^2 y_m^2$$
$$= \frac{1}{2} \times (0.525) \times (9.26) \times (754)^2 \times (0.0085)^2$$
$$= 99.843$$
$$\approx 100 \text{ W}$$

(Ans:)

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17/

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when $y = +2$

$t_2 = 6 \sin(kx + 600t_1 + \phi)$

when $y = -2$

$-2 = 6 \sin(kx + 600 + 2 + \phi)$

From (1) $\longrightarrow$

$\sin(kx + 600t_1 + \phi) = \frac{1}{3}$

$\Rightarrow kx + 600t_1 + \phi = \frac{1}{3} \sin^{-1}\left(\frac{1}{3}\right)$

$\Rightarrow kx + 600t_1 + \phi = 0.34$

From (11) $\longrightarrow$

$kx + 600t_2 + \phi = -0.34$

$kx + 600t_1 + \phi = 0.34$

$600(t_2 - t_1) = -0.68$

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$$\Rightarrow t_2 - t_1 = -\frac{0.68}{600}$$

$$\Rightarrow t_2 - t_1 = -1.13 \times 10^{-3}$$

$$\therefore t_2 - t_1 = -1.13 \text{ ms}.$$

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18 Given, $\omega = 110$ rad/s

$\lambda = 1.80$ m

(a) The angular wave number,

$$K = \frac{2\pi}{\lambda}$$

$$= \frac{2\pi}{1.80} = 3.49 \text{ m}^{-1}$$

(b) The speed of the wave,

$$v = f\lambda$$

$$= \frac{\lambda\omega}{2\pi}$$

$$= \frac{1.8 \times 110}{2\pi} = 31.51 \text{ m/s}$$ (Ans:)

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(a) $T = 4 \cdot 1$

$= 4 \times 0.17$

$= 0.68$ sec. (Ans.)

(b) $f = \frac{1}{T}$

$= \frac{1}{0.68}$

$= 1.470$ (Ans.)

(c) $v = \lambda f$

$= 1.4 \times 1.47$

$= 2.06\ ms^{-1}$. (Ans.)

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(20) (a) $\omega = \frac{v_m}{y_m}$

$= \frac{16}{4 \times 10^{-2}}$

$= 400$ rad/s

$\therefore f = \frac{\omega}{2\pi}$

$\Rightarrow f = \frac{400}{2 \times 3.1416}$

$= 63.66$ Hz (Ans)

(b) Given, $v = 80$ m/s

$\therefore \lambda = \frac{v}{f} = \frac{80}{63.66} = 1.25 \text{ m s}^{-1}$ (Ans)

(c) Given eqn $\Rightarrow y(x,t) = y_m \sin(kx \pm \omega t + \phi)$

Substituting by $y = 4$ cm at $x = 0$, $t = 0$

$\therefore y_m = 0.04$ m (Ans)

(d) We know, $k = \frac{2\pi}{\lambda}$

$= \frac{2\pi}{1.25} = 5$ rad/s

(Ans)

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(e) The angular velocity is given by the relation:

$$\omega = \frac{v_m}{y_m} \Rightarrow \omega = \frac{16}{0.04} = \boxed{400 \text{ rad/s}}$$

(Ans)

(f) Given, $y = 4$ cm at $x = 0, t = 0$ therefore,

$$0.04 = 0.04 \sin(5(0) \pm 400(t) + \phi)$$

$$\Rightarrow 1 = \sin(\phi)$$

$$\Rightarrow \phi = \sin^{-1}(1)$$

$$\therefore \phi = \pi/2$$

$$\therefore Y = 0.04 \sin\left(5x \pm 400(t) + \frac{\pi}{2}\right)$$ (Ans)

(g) The correct choice of sign infront of ω is negative(−) because the wave is shown travelling to positive direction of x-axis.

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